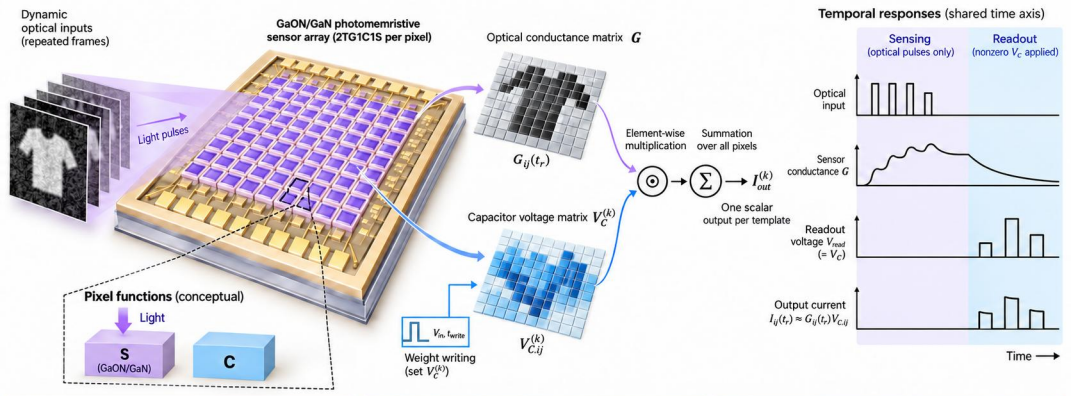
Optical history meets voltage-weighted computing

Schematic



Sensing: external sensor bias = 0; optical history → volatile G



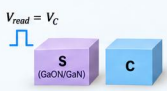
- External sensor bias = 0
- Light pulses accumulate G
- G decays in dark

Weight write: TG1 ON; TG2 OFF; fixed V_{in}, variable t_{write} → stored V_C



- Fixed V_{in} with variable t_{write} sets stored V_C
- Does not reprogram G

Readout: TG1 OFF; TG2 ON; apply nonzero V_C; $I \approx G\,V_C$



- Apply nonzero V_C to S
- $I_{ij}(t_r) \approx G_{ij}(t_r)\,V_{C,ij}$
- Approximately ohmic short readout and small capacitor droop